🦋 Evolution in Action

Name: Class: Date:

Activity 1 · Hands-on · 🏫 In class · ⏱ 30 minutes

🦋 Peppered Moth Hunt

> **You will learn:** Natural selection: prey that match their background survive and leave more offspring.

YOU NEED

- 20 white and 20 dark paper "moths" (small cut-outs)
- A white sheet and a newspaper/dark sheet
- A timer
- A friend to be the "bird"

STEPS

1. Scatter 10 white and 10 dark moths on the white sheet.
2. The "bird" picks up as many moths as possible in 15 seconds.
3. Count the survivors of each colour.
4. Repeat on the dark sheet (polluted bark).
5. Repeat each round 3 times and average your results.

OBSERVATION TABLE

Background	White survived	Dark survived	Which colour was favoured?

QUESTIONS

1. Which moth survived better on the dark background? Why?

2. Is this an example of stabilising, directional or disruptive selection?

3. Give one modern example of evolution caused by humans.

WHAT IS HAPPENING? — THE SCIENCE

Birds catch the moths they can see. On light bark white moths survive; on soot-darkened bark dark moths survive and reproduce. This is how industrial melanism changed moth populations in England after 1850 — natural selection in action.

Key terms: **natural selection** **industrial melanism** **fitness**

Activity 2 · Hands-on · 🏠 At home · ⏱ 40 minutes

✋ Homologous Hands

> **You will learn:** Homologous organs have the same structure but different functions (divergent evolution); analogous organs have different structure but the same function (convergent evolution).

YOU NEED

- Textbook pictures of forelimbs: human, whale, bat, cheetah
- Pictures of a bird wing and a butterfly wing
- Colour pencils

STEPS

1. Draw the bones of the four forelimbs side by side.
2. Colour the same bone in the same colour in every limb (humerus red, radius–ulna blue, carpals green, metacarpals and phalanges yellow).
3. Write the function of each limb.
4. Now draw a bird wing and a butterfly wing: do they have the same bones?
5. Classify each pair as homologous or analogous.

OBSERVATION TABLE

Organ pair	Same structure?	Same function?	Homologous or analogous

QUESTIONS

1. Give one plant example of homologous organs.

2. Are the eyes of octopus and mammals homologous or analogous?

3. What does homology tell us about ancestry?

WHAT IS HAPPENING? — THE SCIENCE

The same colour pattern in all four forelimbs shows a common ancestor — they are homologous, formed by divergent evolution. Bird and butterfly wings both fly but are built differently — analogous, formed by convergent evolution.

Key terms: **homologous organs** **analogous organs** **divergent evolution** **convergent evolution**

Digital activity

💻 Hardy–Weinberg Calculator

> Move the slider to set the frequency of allele A (p) and see the genotype frequencies.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch6/2**

> **You will learn:** In a population in equilibrium: p + q = 1 and $p^2 + 2pq + q^2 = 1$. A change in these frequencies means evolution is happening.

THINK & WRITE

1. If q^2 (aa) = 0.16, what are q and p?

2. Name three factors that disturb Hardy–Weinberg equilibrium.